

INDIVIDUAL ASSIGNMENT

TECHNOLOGY PARK MALAYSIA

CT115-3-3-DLI

DEEP LEARNING FOR INTRUSION DETECTION

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Paper 1: Enhancing ransomware defense: deep learning-based detection and family-wise classification of evolving threats (2024).

Abstract

Ransomware is a type of malware that locks access to or encrypts its victim's files for a ransom to be paid to get back locked or encrypted data. With the invention of obfuscation techniques, it became difficult to detect its new variants. Identifying the exact malware category and family can help to prepare for possible attacks. Traditional machine learning-based approaches failed to detect and classify advanced obfuscated ransomware variants using existing pattern-matching and signature-based detection techniques. Deep learning-based approaches have proven helpful in both detection and classification by analyzing obfuscated ransomware deeply. Researchers have contributed mainly to detection and minimally to family attribution. This research aims to address all these multi-class classification problems by leveraging the power of deep learning. We have proposed a novel group normalization-based bidirectional long-term memory (GN-BiLSTM) method to detect and classify ransomware variants with high accuracy. To validate the technique, five other deep learning models are also trained on the CIC-MalMem-2022, an obfuscated malware dataset. The proposed approach outperformed with an accuracy of 99.99% in detection, 85.48% in category wise classification, and 74.65% in the identification of ransomware families. To verify its effectiveness, models are also trained on 10,876 self-collected latest samples of 26 malware families and the proposed model has achieved 99.20% accuracy in detecting malware, 97.44% in classifying its category, and 96.23% in identifying its family. Our proposed approach has proven the best for detecting new variants of ransomware with high accuracy and can be implemented in real-world applications of ransomware detection.

Hussain, A., Saadia, A., Alhussein, M., Gul, A., & Aurangzeb, K. (2024). *Enhancing ransomware defense: deep learning-based detection and family-wise classification of evolving threats*. *PeerJ Computer Science, 10*, e2546. https://doi.org/10.7717/peerj-cs.2546

Paper 2: Novel Ransomware Detection Exploiting Uncertainty and Calibration Quality Measures Using Deep Learning (2024)

Abstract

Ransomware poses a significant threat by encrypting files or systems demanding a ransom be paid. Early detection is essential to mitigate its impact. This paper presents an Uncertainty Aware Dynamic Early Stopping (UA-DES) technique for optimizing Deep Belief Networks (DBNs) in ransomware detection. UA-DES leverages Bayesian methods, dropout techniques, and an active learning framework to dynamically adjust the number of epochs during the training of the detection model, preventing overfitting while enhancing model accuracy and reliability. Our solution takes a set of Application Programming Interfaces (APIs), representing ransomware behavior as input we call "UA-DES-DBN". The method incorporates uncertainty and calibration quality measures, optimizing the training process for better more accurate ransomware detection. Experiments demonstrate the effectiveness of UA-DES-DBN compared to more conventional models. The proposed model improved accuracy from 94% to 98% across various input sizes, surpassing other models. UA-DES DBNalso decreased the false positive rate from 0.18 to 0.10, making it more useful in real-world cybersecurity applications.

Gazzan, M., & Sheldon, F. T. (2024). *Novel ransomware detection exploiting uncertainty and calibration quality measures using deep learning*. *Information, 15*(5), 262. https://doi.org/10.3390/info15050262

References

Hussain, A., Saadia, A., Alhussein, M., Gul, A., & Aurangzeb, K. (2024). *Enhancing ransomware defense: deep learning-based detection and family-wise classification of evolving threats*. *PeerJ Computer Science, 10*, e2546. https://doi.org/10.7717/peerj-cs.2546

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